

CHE 201 Fall 2019 Midterm

Jordan Eckhart

TOTAL POINTS

40 / 100

QUESTION 1

20 pts

1.1 a 2 / 10

- 0 pts Correct
- ✓ - 2 pts first one is wrong
- ✓ - 2 pts second one is wrong
- 2 pts third one is wrong
- ✓ - 2 pts fourth one is wrong
- ✓ - 2 pts fifth one is wrong

1.2 b 1 / 5

- 0 pts Correct
- ✓ - 2 pts gage pressure is not converted to absolute pressure
- ✓ - 2 pts T is not converted to K
- 5 pts Not done or completely wrong
- 3 pts only pressure is calculated

1.3 c 5 / 5

- ✓ - 0 pts Correct
- 5 pts incorrect

QUESTION 2

2 50 pts

2.1 a 3 / 3

- ✓ - 0 pts Correct

2.2 b 3 / 8

- 0 pts Correct
- ✓ - 1 pts PV state 2 is wrong
- ✓ - 1 pts PV state 3 is wrong
- 1 pts TV state 2 is wrong
- ✓ - 1 pts TV state 3 is wrong
- 8 pts incorrect

✓ - 1 pts PV state 1 is wrong

✓ - 1 pts TV state 1 is wrong

- 4 pts PV graph is wrong

2.3 c 3 / 5

- 0 pts Correct
- 1 pts work expansion compression
- 2 pts KE and PE is zero
- ✓ - 2 pts closed system is not listed

2.4 d 2 / 2

- ✓ - 0 pts Correct
- 1 pts delta E equation is not written
- 1 pts not simplified

2.5 e 7 / 15

- 0 pts Correct
- 3 pts calculation error
- ✓ - 3 pts v3 is wrong
- 3 pts v2 is wrong
- 15 pts incorrect
- 3 pts P is wrong
- ✓ - 3 pts unit conversion
- 3 pts W12 is zero
- ✓ - 2 pts not multiplied with mass

2.6 f 9 / 15

- 0 pts Correct
- ✓ - 3 pts u3 is wrong
- ✓ - 3 pts u1 is wrong
- 2 pts not multiplied with mass
- 2 pts x value is wrong
- 2 pts calculation error
- 15 pts incorrect
- 3 pts q not calculated
- 3 pts not clear calculation

2.7 g 0 / 2

- 0 pts Correct
- ✓ - 0.5 pts M3v not calculated /wrong
- ✓ - 0.5 pts m2v not calculated/ wrong
- ✓ - 1 pts m1v not calculated / wrong

✓ - 10 pts incorrect or not done

- 2 pts calculation error

QUESTION 3

3 30 pts

3.1 a 2 / 2

- ✓ - 0 pts Correct
- 2 pts not done
- 1 pts boundary not labeled

3.2 b 1 / 2

- 0 pts Correct
- 0.5 pts Delta KE and PE is zero not listed
- 1 pts ideal gas is not listed
- ✓ - 1 pts closed system is not listed

3.3 C 2 / 3

- 0 pts Correct
- ✓ - 1 pts Z is not listed or wrong
- 3 pts not done
- 2 pts delta E equation is not listed
- 1 pts not simplified

3.4 d 0 / 3

- 0 pts Correct
- ✓ - 3 pts wrong
- 1 pts slope needs to be negative
- 1 pts Not curved, plotted as linear

3.5 e 0 / 10

- 0 pts Correct
- ✓ - 10 pts not correct or not done
- 6 pts error with integration
- 4 pts not clear explanation
- 2 pts calculation error

3.6 f 0 / 10

- 0 pts Correct

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Please fill in answers in the space supplied and show all your work to receive full credit.
Useful Tables at end of Exam.

1. a) (10 pts) State whether the following statements are **True or False**

T An ideal gas at a given state expands to a fixed final volume at constant pressure or at constant temperature. Work is greater in constant temperature case.

T A two-phase liquid-vapor mixture with equal volumes of saturated vapor and saturated liquid has a quality of 0.5.

F As pressure decreases from the critical pressure, the values of the specific volume of saturated vapor and saturated liquid approach each other.

T If a system undergoes a polytropic process, the system must consist of an ideal gas.

F It takes more energy to vaporize 1kg of saturated liquid water at 100°C than it would at 120°C.

As pressure decreases, the saturated liquid value decreases as the saturated vapor increases

- b) (5 pts) The gage pressure of an automobile tire is measured to be 210kPa before a trip and 220kPa after the trip at a location where the atmospheric pressure is 95kPa. Assuming the volume of the tire remains constant and the air temperature before the trip is 25°C, determine the air temperature in the tire after the trip. (Air is ideal gas)

$$\text{gage pressure}_{\text{before}} = 210 \text{ kPa}$$

$$\text{gage pressure}_{\text{after}} = 220 \text{ kPa}$$

$$\text{atm pressure} = 95 \text{ kPa}$$

$$\text{air temp before} = 25^\circ\text{C}$$

$$\frac{T_2}{T_1} = \frac{P_2}{P_1}$$

$$\frac{T_2}{25} = \frac{220}{210}$$

$$T_2 = 26.19^\circ\text{C}$$

- c) (5pts) The specific volume of saturated liquid water at 0.01°C is $1.002 \times 10^{-3} \text{ m}^3/\text{kg}$. What would be the specific volume of saturated solid at the same temperature, greater or less than saturated liquid value? Why?

$$0.01^\circ\text{C} \quad V_f = 1.002 \times 10^{-3} \text{ m}^3/\text{kg}$$

$$V_g = 206.136 \text{ m}^3/\text{kg}$$

b/c
The value would be greater^{b/c} the solution is solidified so there is more volume to measure. The saturated vapor volume is bigger so the saturated solid volume would be even larger.

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2. (50 pts) 2kg of liquid-vapor mixture of water contained in a piston-cylinder assembly undergoes two processes in series from an initial state where the temperature is 120°C.

- **Process 1-2** System is heated at constant volume to a saturated vapor state at 200°C.
- **Process 2-3** System is compressed at constant pressure until it reaches to 150°C.

Kinetic and potential effects can be ignored.

- (3 pts) Draw a schematic of the system and label the system boundary.
- (8 pts) Carefully sketch the process on a:
 - P-v diagram (Pressure-specific volume)
 - T-v diagram (Temperature-specific volume)

c) (5 pts) State the engineering model

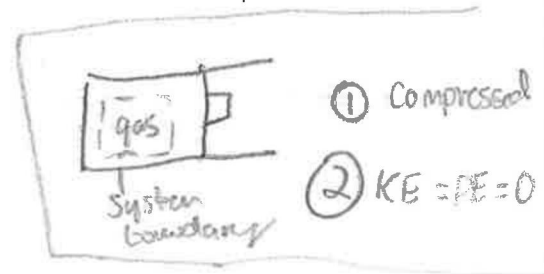
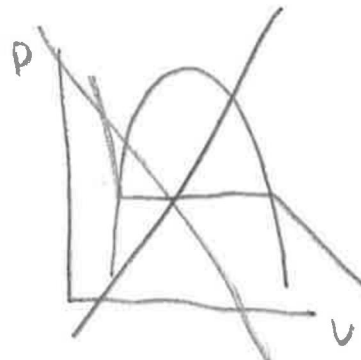
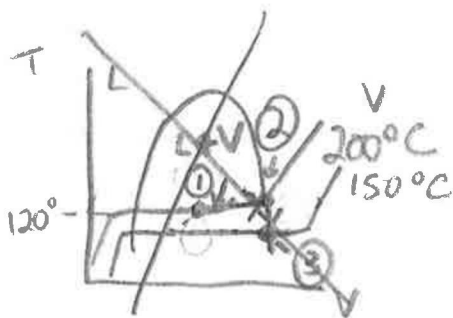
d) (2 pts) Write the simplified energy balance

e) (15 pts) Calculate the work transferred in kJ. Is work done on the system or by the system?

f) (15 pts) Calculate the heat transferred in kJ. Is heat transferred to the system or from the system?

g) (2 pts) Calculate the mass of vapor in each state in kg.

For
T-v and
see P-v
the
next
page



$$\cancel{\Delta KE} + \cancel{\Delta PE} + \boxed{\Delta U = Q - W}$$

$$\Delta U = m(u_2 - u_1)$$

$$2(0.3926 - 1.673)$$

$$= -2.561 = Q + (-15.99)$$

$$\boxed{Q = -18.551 \text{ kJ}}$$

from the system

$$W_{12} = \int p dv = 0 \leftarrow \text{constant volume}$$

$$\boxed{W_{23} = -15.99 \text{ kJ}}$$

work done on system

$$m_{12} = \frac{m_g}{2} = \boxed{0.1}$$

$$T = 200^\circ\text{C}$$

$$P = 15.54$$

$$W_{23} = \int p dv$$

$$= 15.54 (v_2 - v_1) = 15.54 (0.1274 - 1.565)$$

$$T_{\text{initial}} = 120^\circ\text{C}$$

$$T_{\text{final}} = 150^\circ\text{C}$$

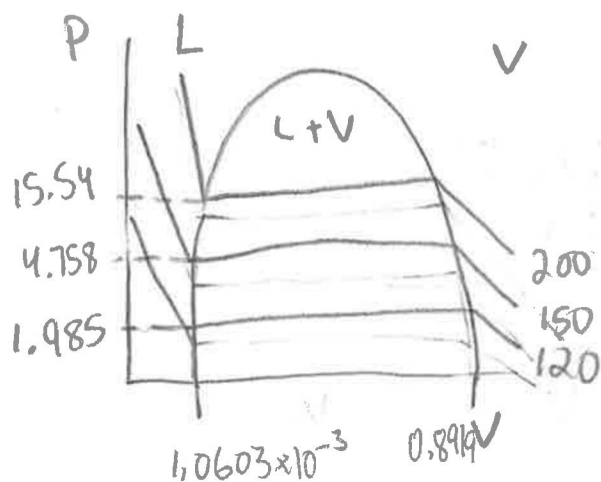
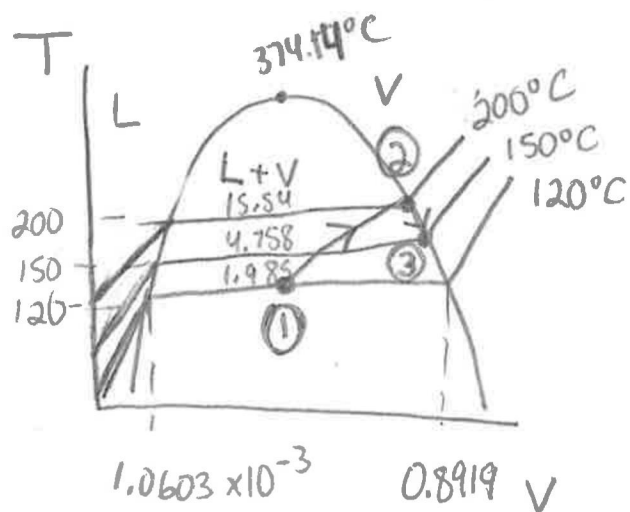
$$P_{\text{initial}} = 1.985 \text{ bar}$$

$$m_{23} = \frac{m_g}{2} = \boxed{0.9}$$

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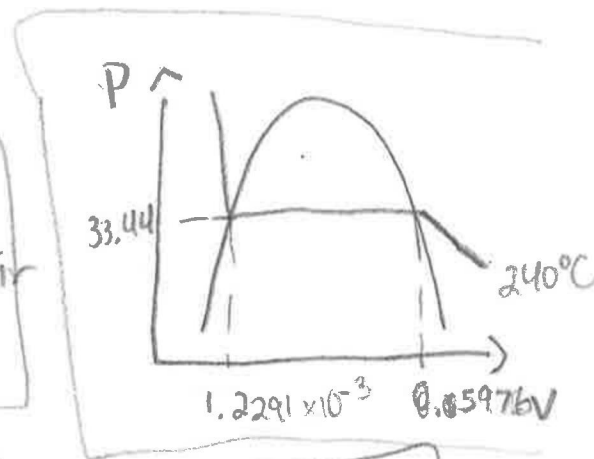
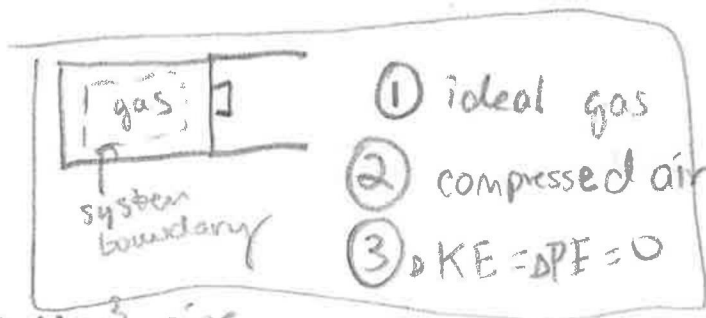
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3. (30 points) A piston cylinder device initially contains 0.4m^3 of air at 100kPa and 240°C . The air is now compressed to 0.1m^3 in such a way that the temperature inside the cylinder remains constant. The air is modeled as an ideal gas and kinetic and potential energy effects are negligible.

- (2 pts) Draw a schematic of the system and label the system boundary.
- (2 pts) State the engineering model
- (3 pts) Write the simplified energy balance and compressibility factor (Z).
- (3pts) Draw P-v diagram (Pressure-specific volume)
- (10 pts) Determine the work (kJ)
- (10 pts) Determine the heat (kJ)



$$V_1 = 0.4\text{m}^3 \text{ air}$$

$$P_1 = 100 \text{ kPa}$$

$$T_1 = 240^\circ\text{C}$$

$$V_2 = 0.1\text{m}^3$$

$$T_2 = 240^\circ\text{C}$$

$$P_2 = 400 \text{ kPa}$$

$$P_2 V_2 = RT$$

$$P_2 (0.1) = 0.287 ($$

$$\Delta \overset{\circ}{KE} + \Delta \overset{\circ}{PE} + \boxed{\Delta U = Q - W}$$

$$Z_{\text{air}} = 0.284$$

$$171.13 = Q - 30$$

$$\boxed{Q = 201.13 \text{ kPa}}$$

$$W = \int P dv$$

$$u_{\text{at } 240} = 171.13 \text{ kJ/kg}$$

$$R = 8.314 \text{ kJ/kmol} \cdot \text{K}$$

$$28.97 \text{ kg/kmol}$$

$$R = 0.287$$

$$W = 100(0.4 - 0.1)$$

$$\boxed{W = 30 \text{ kPa}}$$

$$P_2 = 400 \text{ kPa}$$

$$P_1 V_1 = P_2 V_2$$

$$100(0.4) = (P_2)(0.1)$$

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